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pyOMA2: A Python module for conducting operational modal analysis

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Summary

Operational Modal Analysis (OMA) has garnered considerable attention from the engineering community in recent years and has established itself as the preferred method for estimating the modal properties of structures in Structural Health Monitoring (SHM), particularly in civil engineering. The key advantage of OMA over Experimental Modal Analysis (EMA) is its ability to derive modal parameters solely from output measurements taken during the structure's normal operation. This makes OMA a more practical and efficient approach, as opposed to the traditional EMA, which requires both input and output data.

Statement of need

`pyOMA2` is the latest and improved version of the pyOMA module, a Python library specifically designed for conducting operational modal analysis. It fully utilises Python's object-oriented capabilities to offer a comprehensive suite of tools for performing OMA.

The module supports both single and multi-setup data measurements, enabling users to handle multiple acquisitions that combine reference and roving sensors. It is user-friendly and provides a range of tools for pre-processing and visualising data. A key enhancement is the introduction of interactive plots, allowing users to select the desired modes directly from algorithm-generated graphs. Additionally, a new feature enables users to define the geometry of tested structures, facilitating the visualisation of mode shapes.

The following algorithm are included in the module: - Frequency domain decomposition (FDD); - Enhanced frequency domain decomposition (EFDD); - Frequency spatial domain decomposition (FSDD); - Reference based covariance driven stochastic subspace identification (SSIcov); - Reference based data driven stochastic subspace identification (SSIdat); - Poly-reference Least Square Frequency Domain (pLSCF);

For the readers seeking a more comprehensive understanding of the algorithms and the underlying theory, a wealth of literature is available for consultation. Key references include works by Brincker (Brincker, Ventura, et al., 2001; Brincker, Zhang, et al., 2001; Brincker & Ventura, 2015), Zhang (Zhang et al., 2010), Peeters (Peeters et al., 2004; Peeters & De Roeck, 1999), Van Overschee and De Moor (Van Overschee & De Moor, 2012), Reynders (Reynders, 2012), Rainieri and Fabbrocino (Rainieri & Fabbrocino, 2014), Amador (Amador & Brincker, 2021), Döhler (Döhler, 2011, pp. dohler2013efficient, dohler2013uncertainty), and others.

The module's reliability and applicability for research purposes have been demonstrated by the authors through various studies, such as (Alaggio et al., 2021; Aloisio et al., 2020; Simoncelli et al., 2023). Additionally, the module has gained traction within the research community, as evidenced by its use in studies by (Abuodeh et al., 2023; Croce et al., 2023; Saharan et al., 2023; Talebi et al., 2023), and others.

Module's structure.

The module is structured into three primary levels: 1. At the first level are the `setup` classes. Users instantiate these classes by providing a data array and the sampling frequency for a single setup scenario, or a list of data arrays and their respective sampling frequencies, and reference indices, for a multi-setup scenario. 2. The second level comprises the `algorithms` classes. Users can instantiate the algorithms they wish to run and then add them to the setup class. 3. The third level contains the `support` classes, which serve as auxiliary components to the first two levels. This level includes various specialized classes: * `result` classes, where outcomes are stored. * `geometry` classes, for storing geometric data. * `run_param` classes, where parameters used for running the algorithms are kept. * Dedicated classes for animating mode shapes and interacting with plots generated by the algorithm classes.

In addition to the levels depicted in the figure, there is a further level not shown, comprised of the set of functions internally called by the class methods. Many of these functions represent an updated version of those available in our previous release, `pyOMA`.

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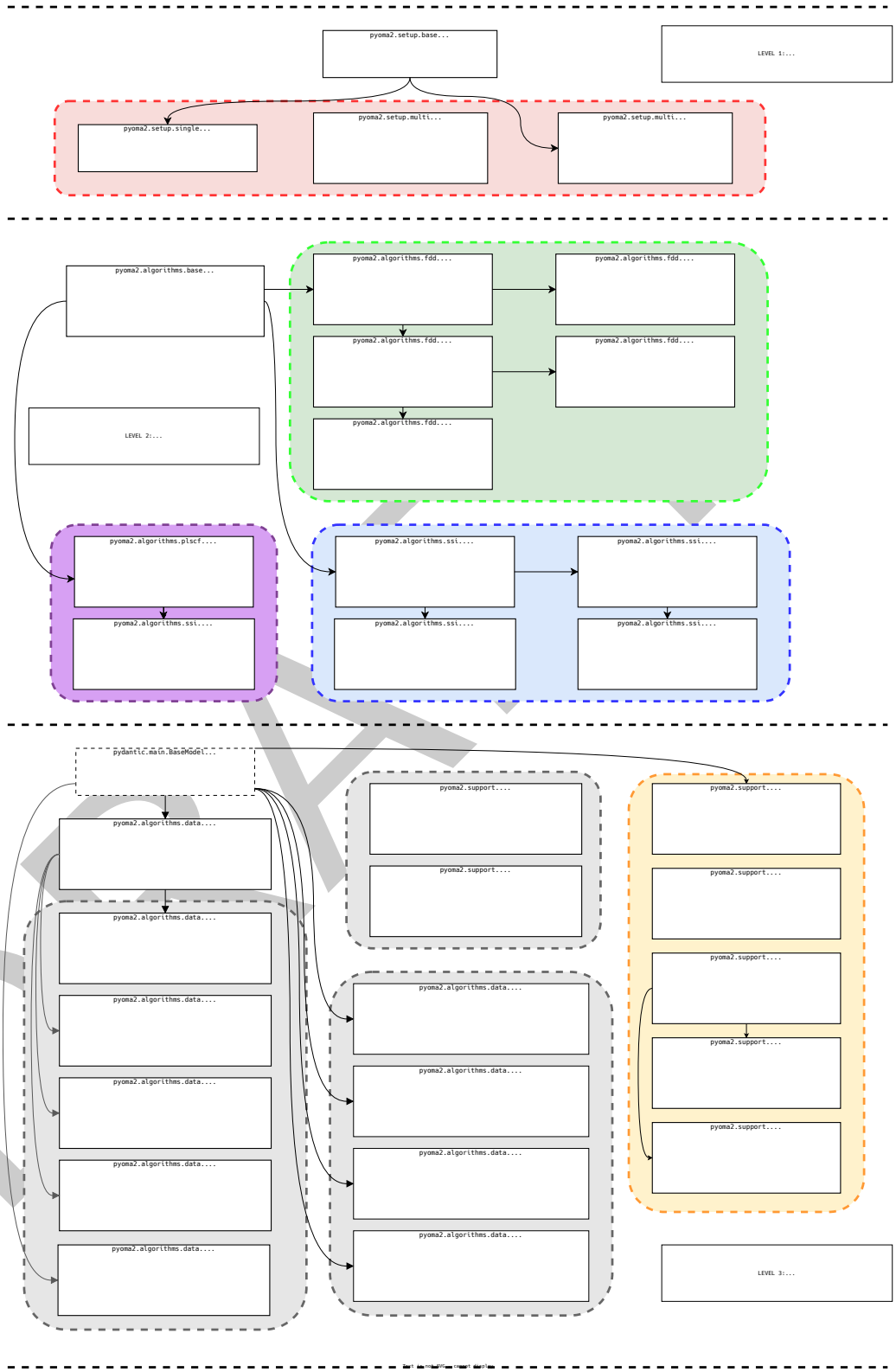

Figure 1: Schematic organisation of the module showing inheritance between classes

Documentation

A comprehensive documentation for `pyOMA2`, including examples, is available at https://pyoma.readthedocs.io/en/main/.

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